Getting started

A practical guide to preparing rainfall data, running an intensity–duration–frequency assessment and producing a traceable technical report.

> **Purpose**
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> The Atlas supports screening and hydrologic analysis. A generated report records the active analysis; it does not certify a design or establish a regulatory return period.

Five-step workflow

1. **Choose data.** Upload a local gauge workbook, retrieve online precipitation, or select a CMIP6 projection workflow.
2. **Confirm location and time coverage.** Check coordinates, spatial extraction and native temporal interval.
3. **Review quality control.** Resolve rejected dates, duplicates, units and incomplete years before interpreting a fit.
4. **Fit and compare.** Select stationary or nonstationary analysis and inspect diagnostics and alternative distributions.
5. **Export an evidence package.** Save the report, Excel calculations and chart together.

Before you begin

Use a modern browser. Online retrieval requires internet access and a reachable data provider. Local gauge processing runs in the browser. Keep a backup of the original workbook and record its source, station name, coordinates and units.

Navigation

Data selects the source; **Map** controls spatial selection; **Analysis** contains fitted curves and diagnostics; **Report** contains the generated document; **Manual** opens this guide.

Prepare and upload gauge data

Input preparation

Use Excel or CSV with a dated precipitation series. Each timestamp should describe one accumulation interval, and rainfall values should be numeric. Clearly identify dates, precipitation, units and whether the native interval is daily, hourly or sub-hourly. Compatible worksheets are scanned automatically; inspect the import summary rather than assuming a particular sheet was selected.

Check	What to do
Dates and times	Use unambiguous dates, preferably ISO format. Confirm timestamp convention and interval boundaries with the source.
Precipitation	Distinguish zero rainfall from missing observations. Do not replace missing values with zero.
Units	Confirm accumulation depth in millimetres or the supported unit selected in the interface. Do not treat a rate as an interval total.
Duplicates	Investigate repeated timestamps in the source. Check rejection and acceptance counts after import.
Location	Enter a meaningful gauge name and coordinates where available. Gauge fitting does not use district boundaries.

After uploading

1. Read the source label, detected native interval, accepted date range and quality-control counts.
2. Compare accepted records with your source workbook.
3. Inspect retained years and annual maxima. At least 90% coverage is required for a retained year.
4. If a result is unavailable, correct the underlying file or select a suitable model; do not fill missing observations solely to enable fitting.

> **Record length**
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> Stationary fitting requires at least eight eligible annual maxima. Fewer than 30 is provisional. Nonstationary fitting requires at least 20 eligible or covariate-matched maxima. These minimums are software eligibility rules, not proof of reliability.

Retrieve online precipitation

Choose the source and native interval

The interface offers ERA5, ECMWF IFS, ERA5-Land, CHIRPS, NASA POWER and supported IMERG workflows. Availability depends on provider, period and interval. CHIRPS is daily; authenticated IMERG half-hourly archive access is not connected. Read the source capability message before retrieval.

ECMWF IFS: hourly or daily precipitation is available from 2017 through the explicit Open-Meteo `ecmwf_ifs` archive. It is an operational model-analysis product, not a homogeneous long-term reanalysis. Model upgrades and its short record require gauge validation and caution for rare return periods. IFS shares Open-Meteo quotas with ERA5. District values use a representative request mesh mapped to model cells and are approximate.

Point or district extraction

1. Select a point/grid-cell series or a supported district spatial series.
2. For a point, verify latitude and longitude. For a district, verify the selected boundary and extraction method.
3. Select the requested dates and interval, then retrieve precipitation.
4. Wait for successful completion. Check returned dates, record count, units, provider and spatial treatment.
5. Inspect the precipitation plot and quality-control summary before fitting.

A gridded point is not a rain gauge. District sampling or polygon statistics are not identical to a single grid cell. Retain the spatial method in any comparison or report.

If retrieval fails

Read the provider error. Try a shorter time window or supported interval when the request is too large. Check coordinate and date limits. If an alternate provider is used, verify that the report identifies the actual dataset and interval. A failed or pending request must not be interpreted as a zero-rainfall record.

> **Evidence to retain**
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> Provider, dataset name, retrieval time, requested and returned period, precipitation units, point coordinates or boundary, spatial extraction method and accepted record count.

Daily versus hourly

Daily input directly supports 24-hour totals. Native hourly data support rolling 1-, 2-, 3-, 6-, 12- and 24-hour durations. Shorter durations require the stated derivation and local validation. A daily series cannot become an observed hourly series by selecting a shorter chart duration.

Fit and interpret IDF curves

Stationary analysis

Select a supported distribution or curve method and compare results. A stationary model holds its parameters constant over time. Review stationarity diagnostics, fit diagnostics, method sensitivity and the number of eligible annual maxima. Failure to reject a trend is not proof that future rainfall will be stationary.

Nonstationary analysis

Choose a time-varying GEV or a supported annual covariate. Confirm at least 20 eligible matched annual maxima. Record the evaluation year or covariate value, whether it is interpolation or extrapolation, the model comparison and the preferred model decision. Correlation alone is not model-selection evidence.

Calculate intensity and understand its constants

Above the chart, enter **t in minutes** and choose return period **T**. Intensity **i in mm/hour** and depth update automatically within the fitted range. Expand “What do the equation and constants mean?” for an explanation. The default Sherman calculator uses `i=a*T^m/(t+b)^n`. It estimates b only with at least three independently supported native durations; daily-derived curves retain an explicitly labelled b=0 simplification. Boundary solutions require review. The reported bounds restrict evaluation. Select **Power · Excel-style trendline** to use `i=c×t^p`: c is a unit-dependent fitted scale and p is the fitted duration exponent. The website regresses ln(i) on ln(t) for each return period, using the active IDF values. R^2 describes that log-space fit and can equal one when duration scaling imposed a power relationship.

Report revision 4.0 includes current c and p values, a worked substitution, fixed versus fitted constant definitions, calculation steps and an acronym list. Appendix A explains formulas; Appendix E defines acronyms. A general formula description does not imply that a method was applied to the active data.

Read the chart and tables

Quantity	Interpretation
Intensity, mm/hour	Mean rainfall rate over the stated duration.
Depth, mm	Intensity × duration in hours. At 24 hours, depth = 24 × intensity.
T-year return level	For a stationary model, annual exceedance probability is 1/T; a 100-year level has 1% AEP.
Nonstationary return level	Conditional on the stated year or covariate value; it is not a timeless recurrence interval.
95% bootstrap interval	Sampling uncertainty under the fitted workflow, not all uncertainty.

Engineering interpretation

Use the governing authority’s criteria to choose the design return period. For a time of concentration calculation, verify flow-path length, slope units and geometric method; a calculated duration does not establish rainfall validity at that duration.

Rare-event caution

The 250- and 500-year levels extend far beyond most available records. Check uncertainty, regional evidence and sensitivity to distribution choice before design use.

CMIP6 projection workflow

Retrieve and correct daily model data

1. Select a GCM, realization/member, SSP, period and spatial request.
2. Retrieve the NASA NEX-GDDP-CMIP6 daily precipitation where the subset service is reachable.
3. Review calendar treatment, precipitation conversion to mm/day and complete model years.
4. Retrieve the historical GCM and the selected gauge or ERA5 reference for the 1985–2014 correction baseline.
5. Apply monthly multiplicative quantile delta mapping (QDM) before extracting future annual maxima.
6. Fit each model separately and review the ensemble of model-specific return levels.

ERA5 correction must be described as reanalysis-corrected. It is not gauge correction. Do not pool model-years as independent observations. Native daily GCM data directly support only 24 hours; observed hourly duration ratios are needed for the supported sub-daily correction workflow.

Correct a district spatial series

Import corrected results when daily retrieval is unavailable

Choose **Import bias-corrected results workbook**. Include **Read Me** rows for Location, Reference and Historical overlap; a **Validation** sheet with result, evidence, severity, impact and required action; and a **Model IDF** sheet with model, member, scenario, period, return period, `Primary_Corrected_24h_Depth_mm` and `Primary_Corrected_Intensity_mm_hr`.

An optional **Ensemble IDF** sheet is reconciled against model rows. At least three models, common return periods, valid coordinates and labels, increasing curves, and consistent depth/intensity values are required. Preserve workbook caveats after import.

> **Interpret ensemble bounds correctly**
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> P10–P90 is inter-model spread, not a 95% sampling-confidence interval. An imported summary supports 24-hour return levels only and does not reconstruct daily records, annual maxima, stationarity tests or alternative distribution fits.

Download and review the report

Available downloads

Format	Use
Print / Save PDF	Open browser printing, select Save as PDF and A4. Use 100% scale; enable background graphics where available. Turn off browser headers/footers if they duplicate the report's own.
Word report (.docx)	A native editable Word document with headings, tables, chart image and page numbering. Pagination may vary between Word and LibreOffice.
Editable HTML	A self-contained document with embedded styles and vector chart. Open in a browser to inspect or print later.
Excel workbook	Calculation and provenance evidence. Retain alongside the report.
User manual	This guide is available in PDF and browser-readable HTML.

Review before sharing

1. Confirm the cover location, source, model, method and issue date.
2. Read the assessment status. A report generated before data loading contains no fitted results.
3. Check the executive decision table against the full IDF table and Excel output.
4. Verify measured versus derived durations, units and the selected evaluation year or covariate.
5. Read quality-control exclusions, uncertainty, limitations and source caveats.
6. Inspect every PDF page for clipped tables, missing charts and isolated headings.
7. Retain the original input and exported evidence. Record any manual edits separately.

Recommended delivery package

Include the PDF report, editable DOCX or HTML, Excel calculation workbook, chart, source-data citation and this manual. Name files with the location, source, scenario where applicable and issue date. Ask an independent reviewer to record their review outside the automatically generated analysis.

Troubleshooting and reproducibility

Symptom	Action
No fitted result	Confirm successful import/retrieval, eligible annual maxima, selected source and model requirements. Re-export after fitting.
Many rejected records	Inspect timestamps, nonnumeric values, negative rainfall, unit selection and duplicate intervals in the original file.
Sub-daily output unavailable	Confirm native interval. Imported CMIP6 summary curves support 24 hours only; use an eligible hourly workflow where appropriate.
Nonstationary fit unavailable	Check at least 20 eligible maxima and matched covariate years. Use the model-status message to identify the missing requirement.
Unexpectedly large return levels	Check units, outliers, shape sensitivity, record length and rare-event extrapolation. Compare supported distributions.
Word export fails	Read the export error, retry in a modern browser, or save HTML/PDF. Allow the download to finish before closing the tab.
PDF table is cut off	Use A4 and 100% scale. Inspect print preview and disable custom browser scaling. HTML retains horizontal scrolling on small screens.

Reproducibility record

Retain the repository commit/version, source workbook, source and retrieval details, selected location and period, QC summary, distribution, stationary/nonstationary setting, covariate evaluation, duration derivation and exports. A report identifier describes the selection; it is not a cryptographic dataset fingerprint.

Native independently corrected sub-daily projections

Select **Native 3-hour projections**, choose an SSP and period, then select the shared district boundary or point coordinates. Click **Retrieve & independently correct native sub-daily curves**. Retrieval can take several minutes; retry resumes completed yearly model subsets in the same tab.

The separate ACCESS-CM2 r1i1p1f1 native CMIP6 3hr precipitation archive is retrieved through ESGF/CEDA, with hourly ERA5 reference data. It supports 3-, 6-, 12- and 24-hour results only. No duration is inferred from daily rainfall. Each duration receives its own historical-model, future-model and reference annual-maxima fit. Correction is $Qcorr(T,d)=Qref,h(T,d) \times Qmodel,f(T,d) / Qmodel,h(T,d)$, followed by $i=Qcorr/(d/60)$. At least 20 common eligible baseline years in 1985–2014 and eight future years are required, using 90% annual coverage.

This corrects fitted return levels, not a continuous sub-daily time series. Bootstrap intervals use 100 year resamples. The native grid is approximately 1.25° × 1.875°; district model aggregation uses polygon overlap, while ERA5 district sampling remains approximate. Reanalysis correction does not establish gauge accuracy. Unsupported durations and retrieval failures are never replaced with scaled daily values. Download native Excel, standalone HTML report or print/save PDF within this workflow. Native point/district results remain separate from the daily CMIP6 workflow.

Developers

The repository is saqibmahmood7117/pakistan-idf-atlas. Use Node.js 22.13 or newer, run `npm ci`, then `npm run dev`. Validate changes with `npm test`, `npm run lint` and `npx tsc --noEmit`. Report exports are implemented in `lib/report-export.ts` and the report section of `app/page.tsx`.

Manual revision 5.0 · November 2026. Interface and provider availability may evolve. The active application’s metadata and validation messages remain the authority for a particular analysis.